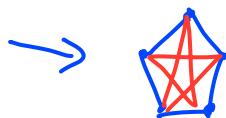


Probability

① Ramsey numbers

$$R(3,3)=6$$

$$\rightarrow K_6 \rightarrow K_3, K_3$$



$$\overset{1989}{\downarrow} 43 \leq R(5,5) \leq \overset{2022}{\downarrow} 48$$

WTS

$$R(100,100) > 2^{50}$$

$$R(6,6) > 8$$

$$R(n,n) > 2^{\frac{n}{2}}$$

Fix 100 vertices

$$\begin{array}{c} A_1 \quad \text{bad} \\ \downarrow \quad \downarrow \\ P(\text{monochromatic}) = \frac{2}{2^{4950}} \end{array}$$

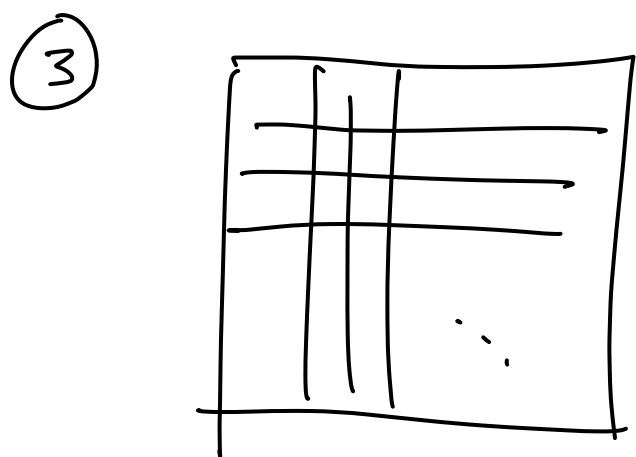
$$P(\text{"bad" exists}) < \frac{2}{2^{4950}} \cdot \binom{2^{50}}{100}$$

$$P(A_1 \cup A_2 \cup A_3 \cup \dots) < \frac{1}{2^{4999}} \frac{(2^{50})^{100}}{100!}$$

2^{50} vertices $\rightarrow$ 2-edge-colouring
no monochromatic K_{100}

$$= \frac{2^{51}}{100!} < 1$$

- ② School $\rightarrow$ 32 clubs, each with 6 members $A_i = i\text{th club violates rule}$
 $P(A_i) = \frac{2}{26} = \frac{1}{13}$
 Some students join Student Union $P(A_1 \cup \dots \cup A_{32}) \leq \frac{1}{13} \times 32 = 1$
 Each club has at least 1, at most 5 representatives in SU
 Prove: this is always possible.



$P(A_1 \cup \dots \cup A_{100})$ 100x100 table
 $\leq \frac{1}{9900} \cdot 5000 < 1$ - 5000: twice



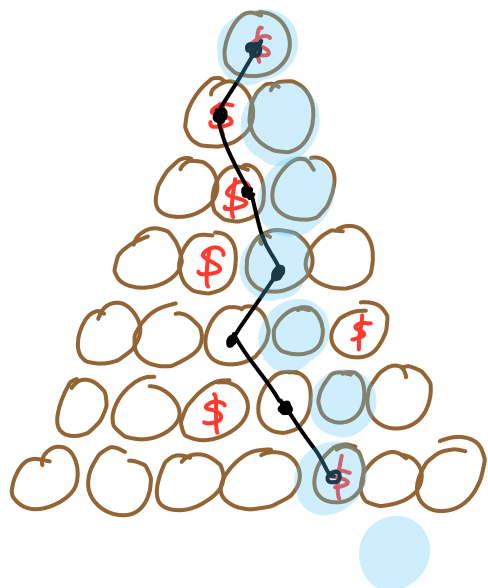
Can place
 Prove: 100 non-attacking rooks
 Covering different numbers
 Random permutation of $(1, 2, \dots, 100)$

$A_i = \text{both } i\text{'s chosen}$

$$P(A_i) = \begin{cases} 0 & \text{if both } i\text{'s in same row/column} \\ \frac{1}{100} \times \frac{1}{99} & \text{otherwise} \end{cases} \leq \frac{1}{9900}$$

Expected value

④ (1 Mo 2023)



Randomly choose $x \in (0, 1)$

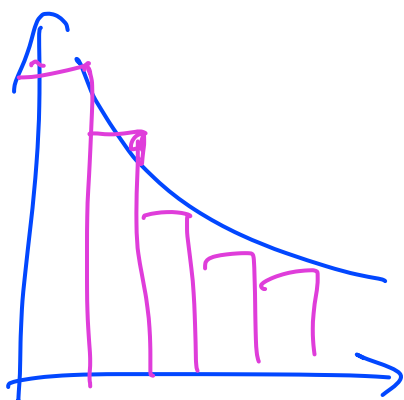
Choose circle $\lceil ix \rceil$ in Row i

e.g. $x=0.7$ 0.7 1.4 2.1 2.8 3.5 4.2 4.9 5.6
 1 2 3 3 4 5 5 6

pass through

$P(\text{"red circle" in Row } i) = \frac{1}{i}$

$$E(\text{no. of red circles}) = \sum_{i=1}^n \frac{1}{i} > \ln n$$



$$\int_1^n \frac{1}{x} dx = \ln n$$

⑤ table tennis tournament
 40 players 80 games played (at most 1 game between 2 players)

(a) Prove: $\exists$ 4 players who have not played

Assume contrary

$$\text{no of games played} \geq \frac{\binom{40}{4}}{\binom{38}{2}} = \frac{40 \cdot 39 \cdot 38 \cdot 37}{38 \cdot 37} \cdot \frac{2}{24} = 130 > 80$$

(b) Prove: $\exists$ 5 players who have not played

Randomly pick each player with prob $p = \frac{1}{4}$

If two picked players have played, discard the loser

$$E(\text{no of players left}) \geq 40 \times \frac{1}{4} - 80 \times \frac{1}{16} = 5$$

$40p - 80p^2$

⑥

a_1	1	2	3	4
a_2	0	9	8	7
			$\vdots$	
a_{100}	2	8	3	3

A = given set of
100 4-digit strings
different

b_1	Choose randomly & independently
b_2	
$\vdots$	
$\vdots$	
b_{100}	

Prove: $\exists B$: set of
100 4-digit strings

s.t.

$$A+B = \{a+b : a \in A, b \in B\}$$

has size ≥ 5000

Claim: For each $i \in [0000, 9999]$, $P(i \in A+B) \geq \frac{1}{2}$

Proof: For fixed i , $P(i \notin A+B) = P(i \notin A+b_1 \cap i \notin A+b_2 \cap \dots \cap i \notin A+b_{100})$

$$= [P(i \notin A+b_1)]^{100}$$

$$= 0.99^{100} = (1-0.01)^{100}$$

$$= 1 - 100 \cdot 0.01 + \underbrace{\binom{100}{2} (0.01)^2}_{4950/10000} \dots$$

Given

WTS

⑦ (APMO 1998) ^(n fixed) $\mathcal{F} = \{ (A_1, A_2, \dots, A_n) : A_i \subseteq \{1, 2, \dots, 1998\} \}$

$\swarrow$ $\swarrow$ $\swarrow$ 2^{1998} choices
 $\nwarrow$ 2^{1998} subsets

Find $\sum_{(A_1, A_2, \dots, A_n) \in \mathcal{F}} |A_1 \cup A_2 \cup \dots \cup A_n|$.

$|\mathcal{F}| = (2^{1998})^n = 2^{1998n}$

Ans: $2^{1998n} \cdot \text{"average"}$
 $= 2^{1998n} \cdot 1998 \cdot \left(1 - \left(\frac{1}{2}\right)^n\right)$

For each $i \in \{1, 2, \dots, 1998\}$

$$\begin{aligned}
 &P(i \in A_1 \cup A_2 \cup \dots \cup A_n) \\
 &= 1 - P(i \notin A_1 \cup A_2 \cup \dots \cup A_n) \\
 &= 1 - \left(\frac{1}{2}\right)^n
 \end{aligned}$$

⑤ table tennis tournament

40 players

80 games played

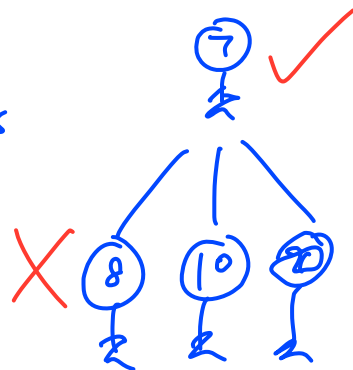
(at most 1 game
between 2 players)

(c) Prove: $\exists$ 8 players who have not played

Randomly label the players 1-40.

Pick a player $\Leftrightarrow$ all his opponents have higher labels

No two picked players have played with each other.



$$E(\text{no of players picked}) = \sum P(i\text{-th player is chosen})$$

$$= \frac{1}{d_1+1} + \frac{1}{d_2+1} + \dots + \frac{1}{d_{40}+1}$$

Player   ... 

no of games played d_1 d_2 ... d_{40}

$$\sum d_i = 160$$

$$\geq \frac{40^2}{(d_1+1) + (d_2+1) + \dots + (d_{40}+1)} = \frac{40^2}{200} = 8$$

A_i : bad event

WTS: $P(A_1 \cup A_2 \cup \dots) < 1$

$$P(A_1 \cup A_2 \cup \dots \cup A_n) < P(A_1) + P(A_2) + \dots$$

$\nwarrow np < 1$

$$\begin{aligned} P(A_1 \cup A_2 \cup \dots) &= 1 - P(\bar{A}_1 \cap \bar{A}_2 \cap \dots) \\ &= 1 - P(\bar{A}_1) \cdot P(\bar{A}_2) \cdot \dots \\ &< 1 \end{aligned}$$

Lovasz Local Lemma

$A_1, A_2, \dots, A_n$: each prob $\leq p$

Each A_i dependent with at most d other A_j 's

If $d \approx \frac{n}{2}$ $e(d+1)p < 1$

If $p(d+1) < \frac{1}{e}$ then $P(A_1 \cup A_2 \cup \dots) < 1$

$d < \frac{n}{3}$

② School $\rightarrow$ ~~32~~^{Some} clubs, each with ≥ 6 members

For each club X ,
members of X join
at most 10 other clubs

Some students join Student Union
(but not all)

Each club has at least 1 ~~at most 5~~ representatives in SU

Prove: this is always possible.

A_i : i -th club violates rule

$$P(A_i) \leq \frac{1}{32} \quad \checkmark$$

Each A_i is dependent with
at most 10 other A_j 's

$$p(d+1) = \frac{1}{32} \cdot 11 < \frac{1}{e}$$

$$P(\text{at least one } A_i) < 1$$

By LL

IMO 2012 Q3

liar's game

⑧ Is it possible to colour each of $1, 2, 3, \dots, 2^{50}$ red or blue such that there is no 100-term monochromatic arithmetic sequence?

$A_1: 1, 2, 3, \dots, 100$ is monochromatic

$A_2: 1, 3, 5, \dots, 199$ is monochromatic

$\vdots$

$$P(A_i \text{ is monochromatic}) = \frac{2}{2^{100}}$$

$$P(A_1 \cup A_2 \cup \dots) \leq \frac{2}{2^{100}} \times \boxed{2^{99}} = 1$$

Exercise Improve the bound 2^{50} to a larger value using LLL.

no of 100-term arithmetic sequences from $1, 2, 3, \dots, 2^{50}$

$$< \binom{2^{50}}{2} = \frac{2^{50}(2^{50}-1)}{2} < 2^{99}$$

$$\underline{12} > \underline{456} \quad _ \quad _ \quad \dots \quad _$$

⑨ School, n clubs, each with ≤ 10 members

VIP = 1
ordinary = 0

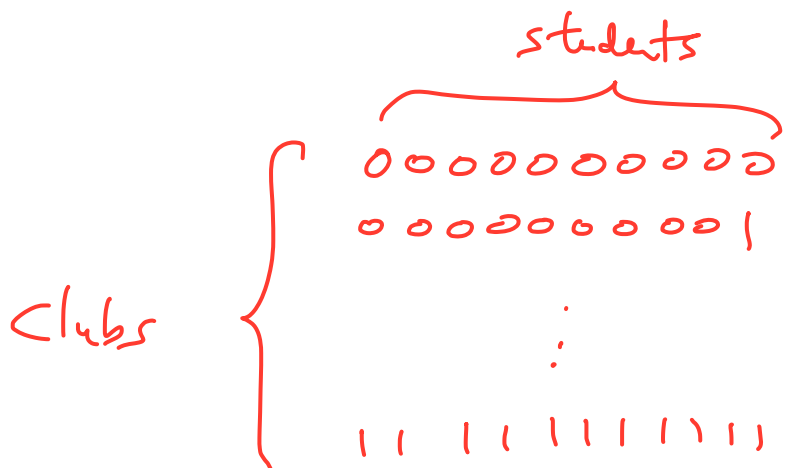
For any 2 different clubs, $\exists$ common member (ordinary in one club
VIP in the other)

Find max n .

student $\begin{cases} \text{nobel} \\ \text{civic} \end{cases}$

A_i and A_j
are mutually
exclusive

A_i = all members of the i -th club
have tiers matching their identity



$$| \geq P(A_1 \cup A_2 \cup \dots \cup A_n)$$

$$= P(A_1) + P(A_2) + \dots + P(A_n)$$

$$\geq \frac{1}{2^{10}} + \frac{1}{2^{10}} + \dots + \frac{1}{2^{10}}$$

$$= \frac{n}{1024}$$

$$\Rightarrow n \leq 1024$$

⑤ table tennis tournament

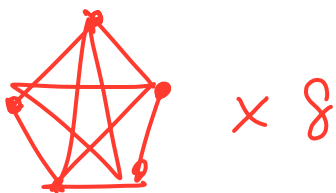
40 players

80 games played

(at most 1 game
between 2 players)

(d) ^{Dis} Prove: $\exists$ 9 players who have not played

8 groups of 5 players each



any 9 players $\rightarrow$ 2 in the same group